Probability

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Contents

Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ is a collection of subsets of Ω. We call $\mathcal{F}$ a *σ-algebra* if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^{\mathsf{c}} \in \mathcal{F}$;
- if $(A_n)_{n\in\mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ-algebra on Ω. A function

$$\mathbb{P} : \mathcal{F} \to [0,1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A.

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space.*

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ-algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω.

Remark. When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω. However, this framework will become useful when considering uncountable Ω.

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4

- *Rolling a fair die.* Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and

$$\mathbb{P}(A) = \frac{|A|}{6}$$

since all outcomes are equally likely.

- $\Omega = \{\omega_1, \ldots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.

- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \ldots, n\}$, and we pick $k \le n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \ldots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.

- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

  What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.

- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \ldots, 9\}^n$, $|\Omega| = 10^n$.

  Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then

$$\mathbb{P}(A_k) = \frac{(k+1)^n}{10^n}$$
$$\implies \mathbb{P}(B_k) = \mathbb{P}(A_k \setminus A_{k-1}) = \frac{(k+1)^n - k^n}{10^n}$$

- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

  Assume each birthday is equally likely to be one of $\{1, \ldots, 365\}$. Then $\Omega = \{1, \ldots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

  Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^{\mathsf{c}} = \{\text{all birthdays different}\}$.

$$\mathbb{P}(A^{\mathsf{c}}) = \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}$$
$$\implies \mathbb{P}(A) = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}$$

  Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \ldots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned}
M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\
&= \frac{n!}{n_1! n_2! \cdots n_k!} \\
&= \binom{n}{n_1, n_2, \ldots, n_k}
\end{aligned}$$

This is the *multinomial coefficient.*

2. Suppose $f : \{1, \ldots, k\} \to \{1, \ldots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

   How many strictly increasing $f : \{1, \ldots, k\} \to \{1, \ldots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.